NHAN VO NGOC THANH

Former fourth-year student, recently graduated in Computer Science (Honor Program)

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DESCRIPTION

- I am a recent graduate of the Honor Program in Computer Science at Ho Chi Minh University of Technology. Passionate about continual learning and problem-solving, I aim to further enhance my expertise in machine learning, natural language processing, and data analysis.
- My objective is to leverage my skills in AI research and data-driven solutions to develop innovative applications that address real-world challenges, ultimately contributing to societal improvement and meaningful impact.

EDUCATION

2021 - 2025 **Ho Chi Minh City University of Technology**

- Honor Program, Bachelor of Computer Science
- Cumulative GPA: 3.7/4.0
- Relevant Coursework: Software Engineering, Database Systems, Data Structures and Algorithms, Computer Architecture, Operating Systems, Computer Networks, Introduction to Artificial Intelligence, Machine Learning, Data Mining, Natural Language Processing
- Interests: Natural Language Processing, Agentic Workflow.

PUBLICATIONS

December 2025 **KGBigData Workshop at IEEE Big Data Conference 2025**

- **Title:** Enhancing Link Prediction with Attentive-HINGE: A Vietnamese Case Study in Educational Knowledge Graph
- **Authors:** Nhan Vo (*), Nhat Truong, Tuan Bui, An Nguyen, Khang Huynh, Tho Quan, Thang Bui.
- **Status**: *Accepted, Camera-Ready Submitted*.
- Will be published in the proceedings of IEEE Big Data 2025.

WORK EXPERIENCES

10/2025 – current **Associate AI Engineer at FPT Software**

- Built an interactive web-based reporting dashboard that queries and visualizes data in real time.
- Integrated AI chatbot agents enabling users to ask insight-driven questions for each dashboard component.
- Designed and implemented the workflow for a dedicated report agent, focusing on how it interprets report metadata and responds to analytical queries.
- Optimized agent prompts and defined rule-based mechanisms to detect and respond to anomalies in report data.

5/2025 – 8/2025 **Game Development Fresher at ZingPlay Studio – VNG Corporation**

- Developed "Chaos Age" – a real-time strategy game (Clash of Clans style).
- Implemented A* pathfinding, opponent matching, and target selection for troops.
- Implement troop training (client), moving, building and upgrades logic (server).
- Synchronized battle results between client and server, ensuring data integrity.

6/2024 - 8/2024 **AI Engineer Intern at Rockship Co.**

- Researched and implemented agentic workflows leveraging LangChain and LlamaIndex for AI product development.
- Built and optimized an AI-powered Retrieval-Augmented Generation (RAG) pipeline integrated with PostgreSQL and PGVector for document retrieval.
- Designed and developed graph-based workflow models for asynchronous execution using LangGraph.
- Delivered successful demos for automation projects, including email invoice parsing and item matching with similarity search.

1/2024 - 6/2024 **Teaching Assistant at Ho Chi Minh City University of Technology**

- Guided students in practical exercises, ensuring understanding of core computer science concepts.
- Designed and evaluated assignments, contributing to the academic growth of over 40 students per semester.
- Created and graded practice exams to prepare students for final assessments.

PROJECTS

10/2025 – current **C99 - Agentic Report** **NDA Contract**

- **Description:** Developed an AI-enhanced reporting web app that queries and visualizes data in real time, enabling users to interact with dashboards through natural-language insights.
- **Technical Skills:** Agentic workflow design, Prompt engineering, Rule-based anomaly detection, FastAPI, Swagger, LangGraph, PostgreSQL.
- **Soft Skills:** Problem-solving, Analytical thinking, Cross-team communication, Rapid prototyping, Requirement clarification.
- **Responsibility:** Designed and implemented the workflow for the report-level agent; optimized prompts for accurate insight generation; defined rules to detect anomalies in user queries and report data; collaborated with the engineering team to integrate the agent with the reporting dashboard.

5/2025 – 8/2025 **Chaos Age – Real-time Strategy Game (Clash of Clans style)** **View showcase**

- **Description:** A tactical strategy game where players build bases, train troops, and attack others.
- **Technical Skills:** Java, Cocos2d-JS, Client-Server Architecture.
- **Soft Skills:** System thinking, problem solving, collaboration.
- **Responsibility:** Implemented troop movement (A*), opponent matching logic, training system (client), and building/upgrading system (server). Ensured reliable match result synchronization.

7/2024 - 8/2024 **Invoice and Purchase Order Automation:** **NDA contract**

- **Description:** Created email parsers for extracting Invoice and Purchase Orders data, reducing manual effort.
- **Technical Skills:** FastAPI, Swagger, LlamaParse, PostgreSQL.
- **Soft Skills:** System thinking, problem solving, collaboration.
- **Responsibility:** Designed similarity search features for matching items between client-provided data and company inventory.

6/2024 - 7/2024 **AISync Workflow Framework** **NDA contract**

- **Description:** Developed modular components to process documents and queries with high efficiency.
- **Technical Skills:** FastAPI, SQLAlchemy, LangGraph, LangFuse, Swagger, PostgreSQL.
- **Soft Skills:** System thinking, problem solving, collaboration.
- **Responsibility:** Implemented a document memory module and used LangFuse for real-time monitoring and debugging.

9/2024 – 5/2025 **Hyper-relational Knowledge Graph Reasoning**

- **Descriptions:** Designed and evaluated an attention-based model (Attentive-HINGE) for improved multi-hop QA over complex knowledge graphs.
- **Technical Skills:** PyTorch, Deep Learning, CNN, Self-Attention, Knowledge Graph, Python
- **Soft-Skills:** Leadership, Presentation skills, Teamwork.
- **Responsibility:** Design and implement Attentive-HINGE (attention-based model base on HINGE -SOTA model for link prediction)

4/2024 – 5/2024 **Traffic Light and Sign Detection**

- **Descriptions:** This component of the Autonomous Driving project focuses on developing fully autonomous driving solutions, such as lane detection and traffic light recognition in the Gazebo simulator, before being deployed to ROS for real-world implementation.
- **Technical Skills:** YOLOv5, Pandas, NumPy, Python
- **Soft-Skills:** Presentation skills, Teamwork.
- **Responsibility:** Conduct data processing and model training with YOLOv5 to perform traffic sign detection.

3/2024 – 4/2024 **Weather forecasting**

- **Descriptions:** Addressing weather forecasting with Machine Learning and Big Data tools (PySpark) to show whether is possible to make valuable predictions of meteorological conditions only based on previously seen meteorological data and giving a set of weather measurements, to predict which meteorological condition should occur
- **Technical Skills:** PySpark, Koalas, Pandas, NumPy, Python
- **Soft-Skills:** Presentation skills, Teamwork.
- **Responsibility:** Implement the ML model (RF, MLP, ...) to predict on preprocessed data

CERTIFICATES

English	**TOEIC - Listening and Reading: 810/990 points**	**March 2024 - March 2026**
Coursera	**Python for Data Science, AI & Development**	**View online certificate**
Coursera	**Databases and SQL for Data Science with Python**	**View online certificate**

ACHIVEMENT

Dec 2025	**Student of 5 Merits Award**	**Vietnam National University, Ho Chi Minh City**
July 2025	**EVN Scholarship**	**Ho Chi Minh University of Technology**

- Awarded by Vietnam Electricity (EVN) for academic excellence and strong ethics.

Nov 2024	**Outstanding Student for the academic year 2023–2024**	**Ho Chi Minh University of Technology**
Dec 2023	**University's merit-based scholarship**	**Ho Chi Minh University of Technology**

- Awarded by Ho Chi Minh University of Technology for academic excellence in top 10% of each faculty.

Nov 2023	**Excellent Student or the academic year 2022–2023**	**Ho Chi Minh University of Technology**